09. Additional Functionality & Value-Added Features

Comprehensive Beyond-SRS Engineering Report • HarvestHub v2.0

EXECUTIVE ENGINEERING SUMMARY & RATIONALE

While the original Software Requirements Specification (SRS) defined a foundational baseline focusing on simulated checkout and basic CRUD catalogs, HarvestHub v2.0 has been elevated into an enterprise-grade agricultural ecosystem. The 23 modular enhancements documented below address critical real-world challenges including multi-vendor dispatch isolation, live GPS mapping, multi-modal internal communications, cryptographic OTP security, and localized AI intelligence.

9.1 Authentication, Cryptographic Security & Account Recovery

AF-01 **Dual-Tier Registration with Automated Email OTP Verification**

Manual sign-ups trigger an automated 6-digit cryptographic, time-limited OTP (SHA-256 hashed) dispatched via custom SMTP gateway. Prevents bot registrations and guarantees validated contact details for authentic transactions.

AF-02 **Seamless Google & Apple One-Tap Social Authentication**

Integrated Google Sign-In with OAuth 2.0 and profile image synchronization. Users logging in via Google bypass manual OTP while automatically creating a verified cloud profile in Firestore.

AF-03 **Self-Service OTP-Based Password Reset & Instant Session Sync**

Enables users to reset forgotten passwords securely via time-bounded OTP verification. New credentials compute a secure hash locally and sync with Firestore, completely eliminating lockouts.

9.2 Real-Time Geolocation & Interactive Map Ecosystem

AF-04 **Cross-Role Interactive Real-Time Farmers & Markets Map**

Integrated Leaflet/GPS map accessible across Customer, Farmer, and Admin panels. Renders live interactive map pins of verified growers, farm gates, roadside stalls, and wholesale mandis.

AF-05 **Farmer Dynamic Market Pin Drop & Operating Hours Configuration**

Farmers can pinpoint exact live market coordinates and daily operational hours (e.g., 7:00 AM - 2:00 PM) directly from their portal, updating customer map views instantaneously.

AF-06 **Hierarchical City/Neighborhood Filtering with 50km GPS Radius**

Allows buyers to discover growers by Pakistani cities (Karachi, Lahore, Islamabad) and specific sub-areas (Clifton, Gulshan, Malir, North Karachi) with real-time GPS distance calculation within 50 km.

AF-07 **1-Tap Turn-by-Turn GPS Navigation Integration**

Directly launches native navigation (Google Maps / Apple Maps) from any farmer profile or map pin, providing seamless driving and walking routes directly to the farm.

Section 09: Additional Functionality (Beyond SRS)

Continuous Feature Specifications & Implementation Details

9.3 Multi-Vendor Order Splitting & Direct Communication

AF-08 **Automatic Multi-Vendor Order Splitting Engine**

When a customer checks out items from multiple distinct farmers in a single cart, the backend automatically partitions the order into isolated sub-orders. Each farmer only receives and fulfills their specific produce.

AF-09 **Granular Sub-Order Tracking with Direct Cellular Calling (tel:)**

Customers track individual sub-order statuses per farmer and can initiate 1-tap direct GSM phone calls to the specific farmer for real-time delivery coordination.

AF-10 **Unified Real-Time Multimedia Communication Suite**

Built-in 1-on-1 messaging connecting customers and farmers: Real-time text chat with unread counters, push-to-talk voice notes recording/player, camera photo sharing, and voice/video calling interfaces.

9.4 Marketplace Intelligence & Produce Discovery

AF-11 **Farmer Transparent Marketplace Price Monitor**

Provides farmers with real-time price monitoring of competing produce across nearby markets, empowering growers to price produce competitively and eliminate predatory middlemen markups.

AF-12 **Multi-Criteria Dynamic Search & Filtering Matrix**

High-speed search across Product Name, Category, Farmer Name, Distance slider, Budget-to-Premium price limits, and Farm-Freshness ratings with instant 1-tap filter reset.

AF-13 **Customer Product Reviews & Multi-Criteria Star Ratings**

Verified buyers post star ratings and text reviews, automatically recomputing aggregate ratings across both individual products and the farmer's public profile.

9.5 Logistics, Pickup Scheduling & Inventory Automation

AF-14 **Farmer Self-Managed Pickup Slot Scheduling Engine**

Farmers configure customizable pickup windows (e.g. 9:00 AM - 11:00 AM) with slot quotas. Customers select pickup slots at checkout to bypass delivery charges, supported by local offline caching.

AF-15 **Social Farmer Following & Restock Broadcast Alerts**

Customers can follow specific farmers. The app triggers automatic in-app alerts and notifications whenever followed farmers harvest fresh crops or restock items.

AF-16 **Live Inventory Guard & Automated Low-Stock Warnings**

Real-time stock depletion prevents overselling of zero-stock items while dispatching proactive low-stock warnings (< 10 units) to farmers.

Section 09: Additional Functionality (Beyond SRS)

Continuous Feature Specifications & Implementation Details

9.6 Gamification, Loyalty Rewards & Social Proof

AF-17 **Customer Eco-Loyalty Tier System & Checkout Discounts**

Tiered loyalty progression (Bronze, Silver, Gold, Eco-Champion) awarding reward points on orders redeemable for instant checkout discounts, paired with a live Customer Leaderboard.

AF-18 **Farmer Performance & Fulfillment Leaderboard**

Live competitive ranking of farmers based on delivery fulfillment speed, quality ratings, and transaction volume to motivate top-tier agricultural standards.

9.7 Localization, Artificial Intelligence & Administration

AF-19 **Native Bilingual Architecture (Urdu & English)**

Full bilingual implementation supporting native Urdu (ur_PK) and English (en_US) across all screens, metrics, bottom navigation, and dialogs with instant toggle and persistence.

AF-20 **HarvestAI Botanical & Live Platform Knowledge Assistant**

Context-aware AI assistant (powered by Gemini API with intelligent local fallback) with real-time knowledge of live farmer counts, cheapest sellers, produce health benefits, and app guides in Urdu & English.

AF-21 **Admin Real-Time Farmer Verification & Badge Management**

Administrators inspect grower credentials and toggle verified trust badges (isVerified), displaying the green verification tick across product cards and public profiles.

AF-22 **Admin Centralized Moderation & Safety Oversight**

Full administrative transparency over platform-wide message safety, customer dispute moderation, user account suspensions, and live Gross Merchandise Value (GMV) metrics.

AF-23 **Rich Motion Graphics & Video Hero Card Integration**

Seamless looping background MP4 video hero banners across Customer, Farmer, and Admin home dashboards for a premium visual experience.

Summary Comparison: Baseline SRS Scope vs. HarvestHub v2.0 Implemented Reality

Module / Domain	Baseline SRS Scope	HarvestHub v2.0 Implemented Reality
Authentication	Basic Email/Password form	Email OTP Verification + Google OAuth 2.0 + Instant Password Hash Recovery
Location & Map	Static text address string	Interactive Real-Time Leaflet Map + Pakistan/Karachi Filter + 50km GPS Radius + Navigation
Communication	Explicitly out of scope	Real-time Chat + Push-to-Talk Voice Notes + Camera Photos + Audio/Video Call UI
Order Processing	Single generic simulated order	Automatic Multi-Vendor Order Splitting + Direct Phone Calling to Specific Farmer
AI Assistant	Basic static FAQ text answers	Live Database-Aware Context AI (Gemini + Local Engine) in Urdu, Roman Urdu & English
Gamification	Not mentioned in SRS	Dual Eco-Reward Leaderboards (Customer Discount Points & Farmer Speed Badges)
Pickup Scheduling	Optional / Basic listing	Farmer Self-Managed Slot Scheduling Engine with Quotas & Offline Cache Sync
Language Support	English only	Full Native Bilingual System (Urdu & English) across all screens & dialogs
Admin Moderation	Standard CRUD table list	Live Conversation Monitoring + Verified Farmer Badges + Platform Safety Reports
Design Aesthetics	Standard mobile templates	Premium Dark Emerald Glassmorphic UI with Looping MP4 Video Hero Cards